Lesson 12 Teacher's Key: Hearing It Happen

TIC-80 - Instructor Reference Only

VIDEO GAME CREATION

What This Lesson Teaches

The plainest possible version of this week's idea: five sfx() calls in five existing branches, using only what Lesson 2's Sound Editor already established - and a chance to explicitly retire the hasWon-style replay-guard flag Lesson 6 needed, now that Lesson 10/11's restructuring made every trigger point naturally one-shot.

Level 1 - Full Worked Solution

- sfx(2) (hit) moves to directly follow the score/enemies_remaining updates inside the bullets loop's hit branch - the same restructured location every other engine's hit sound now lives in.
- sfx(4) (wave) and sfx(5) (lose) are the two genuinely new sounds, placed exactly where Lesson 11's nested if and the enemy escape-check already live.
- sfx(3) (win), if a student built it back in Lesson 6, may originally have needed a hasWon guard flag to avoid replaying every frame - as of Lesson 10/11's restructuring into a one-shot hit branch, that guard is no longer necessary and can be removed if still present.

Why This Structure

- TIC-80's collision detection scans every frame, but a hit only ever causes enemies_remaining to cross to a new value once - so code placed directly inside that specific crossing (not a separately-polled check in TIC() itself) is naturally edge-triggered, with no manual guard needed.
- This is a good moment to point out that Lesson 6's original hasWon-flag workaround was a reasonable fix for the code structure at the time - and that Lesson 10's game_active refactor incidentally solved the same problem more thoroughly, a nice example of good structure paying dividends beyond its original purpose.

Level 2 / Level 3 Talking Points

- Ask students who built Lesson 6's optional win sound to check whether they still have a hasWon guard flag anywhere, and if so, to trace through whether it's still doing anything useful post-Lesson-10 - a good structural-reasoning exercise, not just a code-reading one.
- For advanced students: ask them to name all five SFX Editor slots and their triggers from memory as a quick comprehension check before moving to lab time.

Common Student Mistakes

- Placing sfx(4) outside the else branch, causing it to also fire on the true win.
- Mixing up slot numbers between 3, 4, and 5 - easy to do since they're consecutive and were designed close together in the SFX Editor.
- Leaving a stale hasWon-guard flag in place from Lesson 6 that now silently prevents the win sound from ever playing a second time across multiple playthroughs without a full restart.